

The reconstructed embeddings from the selected features were passed through the ESM2 MLP layer to predict logits for the 20 amino acids, which were then converted into LLRs between the mutant and wild-type amino acids (equation 8). The resulting LLRs were compared with experimentally measured mutation effects on protein stability “ddG_ML” to compute a soft Spearman correlation loss:

$$L = -\frac{\text{Cov}(r(\text{ddG}), r(\text{LLR}))}{\sqrt{\text{Var}(r(\text{ddG}))}\sqrt{\text{Var}(r(\text{LLR}))}} + \lambda|g| \quad (11)$$

where $r(\cdot)$ denotes the differentiable rank function⁵¹. The first term maximizes the rank correlation between predicted and experimental mutation effects, while the second imposes an L1 regularization on the gate values to constrain the number of active features. The weighting coefficient λ was set to 1 based on performance on evaluation set.

Model training was implemented in PyTorch using the Adam optimizer (learning rate = 1×10^{-3} , default parameters). Each batch contained a single protein, and gradients were accumulated across all training proteins before updating the gate parameters, meaning the parameters were updated once per epoch. Models were trained for 60 epochs, and the checkpoint with the highest validation Spearman correlation after 50 epochs was selected for downstream analyses.

MotifAE-G protein design and stability prediction

For each representative test protein, we performed four rounds of iterative design. In each round, the probabilities of all possible single substitutions were predicted using ESM2 and MotifAE-G, with different weights applied to stability-associated features. The wild-type sequence was used without masking, following the same procedure as in mutation effect prediction. All possible mutations were ranked by their relative probability compared to the wild type: $p(x_i^{mt} | x) / p(x_i^{wt} | x)$. A top-k sampling strategy³⁷ (with $k=10$ in our experiments) was applied to select mutations, where the sampling probability of each mutation was proportional to its relative probability.

To evaluate the stability of the designed sequences, we predicted their structures using ESMFold. Global stability was then estimated using FoldX (version 20251231) based on the predicted structures. Specifically, we first ran the command “RepairPDB”, followed by “Stability” to compute energy contributions, and used the sum (total energy) as an absolute stability metric. We reversed the sign of this total energy to represent ΔG . Only designs with ESMFold pLDDT > 0.8 and FoldX-predicted $\Delta G > 2$ kcal/mol were analyzed.

Data availability

All data used in this work are publicly available. 2.3 million representative proteins were downloaded from <https://afdb-cluster.steineggerlab.workers.dev/>. ProteinGYM DMS data were downloaded from <https://proteingym.org/download>. ELM motifs were downloaded from <http://elm.eu.org/downloads.html>. For protein stability, the DMS data and AlphaFold2-predicted protein structures were downloaded from <https://zenodo.org/records/7992926>.

Code availability

Codes are available at GitHub: <https://github.com/CHAOHOU-97/MotifAE.git>.

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Contributions

Y.S. and C.H. conceived the study. C.H. designed and implemented the experiments. D.L. assisted with analyses of homodimers and mutations. All authors evaluated and interpreted the results and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

References

1. Gribskov, M., McLachlan, A. D. & Eisenberg, D. Profile analysis: detection of distantly related proteins. *Proceedings of the National Academy of Sciences* **84**, 4355–4358 (1987).
2. Kumar, M. *et al.* ELM-the Eukaryotic Linear Motif resource-2024 update. *Nucleic Acids Res* **52**, D442–D455 (2024).
3. Aditham, A. K., Markin, C. J., Mokhtari, D. A., DelRosso, N. & Fordyce, P. M. High-Throughput Affinity Measurements of Transcription Factor and DNA Mutations Reveal Affinity and Specificity Determinants. *Cell Systems* **12**, 112-127.e11 (2021).
4. Örd, M. *et al.* High-throughput investigation of cyclin docking interactions reveals the complexity of motif binding determinants. *Nat Commun* **16**, 7622 (2025).
5. Koren, I. *et al.* The Eukaryotic Proteome Is Shaped by E3 Ubiquitin Ligases Targeting C-Terminal Degrons. *Cell* **173**, 1622-1635 e14 (2018).
6. Zhang, Z. *et al.* Elucidation of E3 ubiquitin ligase specificity through proteome-wide internal degron mapping. *Molecular Cell* **83**, 3377-3392.e6 (2023).
7. Tokheim, C. *et al.* Systematic characterization of mutations altering protein degradation in human cancers. *Mol Cell* **81**, 1292-1308 e11 (2021).
8. Savojardo, C., Martelli, P. L. & Casadio, R. Finding functional motifs in protein sequences with deep learning and natural language models. *Current Opinion in Structural Biology* **81**, 102641 (2023).
9. Hou, C., Li, Y., Wang, M., Wu, H. & Li, T. Systematic prediction of degrons and E3 ubiquitin ligase binding via deep learning. *BMC Biol* **20**, 162 (2022).
10. Lin, Z. *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130 (2023).
11. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* <https://doi.org/10.1038/s41586-021-03819-2> (2021) doi:10.1038/s41586-021-03819-2.
12. Kulmanov, M. *et al.* Protein function prediction as approximate semantic entailment. *Nature Machine Intelligence* **6**, 220–228 (2024).
13. Zhang, Z. *et al.* Protein language models learn evolutionary statistics of interacting sequence motifs. *Proc Natl Acad Sci U S A* **121**, e2406285121 (2024).
14. Vig, J. *et al.* Bertology meets biology: Interpreting attention in protein language models. (2020).
15. Zhang, Y., Zheng, J. & Zhang, B. Protein Language Model Identifies Disordered, Conserved Motifs Implicated in Phase Separation. *eLife* **14**, (2025).
16. Cunningham, H., Ewart, A., Riggs, L., Huben, R. & Sharkey, L. Sparse Autoencoders Find Highly Interpretable Features in Language Models. Preprint at <https://doi.org/10.48550/arXiv.2309.08600> (2023).
17. Yun, Z., Chen, Y., Olshausen, B. A. & LeCun, Y. Transformer visualization via dictionary learning: contextualized embedding as a linear superposition of transformer factors. Preprint at <https://doi.org/10.48550/arXiv.2103.15949> (2023).
18. Brix, G. *et al.* Genome modeling and design across all domains of life with Evo 2. 2025.02.18.638918 Preprint at <https://doi.org/10.1101/2025.02.18.638918> (2025).
19. Adams, E., Bai, L., Lee, M., Yu, Y. & AlQuraishi, M. From Mechanistic Interpretability to Mechanistic Biology: Training, Evaluating, and Interpreting Sparse Autoencoders on Protein Language Models. 2025.02.06.636901 Preprint at <https://doi.org/10.1101/2025.02.06.636901> (2025).
20. Simon, E. & Zou, J. InterPLM: discovering interpretable features in protein language models via sparse autoencoders. *Nat Methods* **22**, 2107–2117 (2025).
21. Gujral, O., Bafna, M., Alm, E. & Berger, B. Sparse autoencoders uncover biologically interpretable features in protein language model representations. *Proceedings of the National Academy of Sciences* **122**, e2506316122 (2025).
22. Brandes, N., Goldman, G., Wang, C. H., Ye, C. J. & Ntranos, V. Genome-wide prediction of disease variant effects with a deep protein language model. *Nat Genet* **55**, 1512–1522 (2023).

23. Notin, P. *et al.* ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. *Advances in Neural Information Processing Systems* **36**, 64331–64379 (2023).
24. Hou, C., Liu, D., Zafar, A. & Shen, Y. Understanding Language Model Scaling on Protein Fitness Prediction. 2025.04.25.650688 Preprint at <https://doi.org/10.1101/2025.04.25.650688> (2025).
25. Barrio-Hernandez, I. *et al.* Clustering predicted structures at the scale of the known protein universe. *Nature* **622**, 637–645 (2023).
26. Landrum, M. J. *et al.* ClinVar: updates to support classifications of both germline and somatic variants. *Nucleic Acids Res* **53**, D1313–D1321 (2025).
27. Okada, M. Regulation of the Src Family Kinases by Csk. *Int J Biol Sci* **8**, 1385–1397 (2012).
28. Erdos, G., Pajkos, M. & Dosztanyi, Z. IUPred3: prediction of protein disorder enhanced with unambiguous experimental annotation and visualization of evolutionary conservation. *Nucleic acids research* **49**, W297–W303 (2021).
29. Kshirsagar, M. *et al.* Rapid and accurate prediction of protein homo-oligomer symmetry using Seq2Symm. *Nat Commun* **16**, 2017 (2025).
30. Kovtun, D. *et al.* PINDER: The protein interaction dataset and evaluation resource. 2024.07.17.603980 Preprint at <https://doi.org/10.1101/2024.07.17.603980> (2024).
31. Andreatta, M., Alvarez, B. & Nielsen, M. GibbsCluster: unsupervised clustering and alignment of peptide sequences. *Nucleic Acids Res* **45**, W458–W463 (2017).
32. Tsuboyama, K. *et al.* Mega-scale experimental analysis of protein folding stability in biology and design. *Nature* **620**, 434–444 (2023).
33. Hou, C., Zhao, H. & Shen, Y. Learning Biophysical Dynamics with Protein Language Models. *bioRxiv* 2024.10.11.617911 (2025) doi:10.1101/2024.10.11.617911.
34. UniProt, C. UniProt: a worldwide hub of protein knowledge. *Nucleic Acids Res* **47**, D506–D515 (2019).
35. Dieckhaus, H., Brocidiaco, M., Randolph, N. Z. & Kuhlman, B. Transfer learning to leverage larger datasets for improved prediction of protein stability changes. *Proceedings of the National Academy of Sciences* **121**, e2314853121 (2024).
36. Madani, A. *et al.* Large language models generate functional protein sequences across diverse families. *Nat Biotechnol* **41**, 1099–1106 (2023).
37. Darmawan, J. T., Gal, Y. & Notin, P. Sampling Protein Language Models for Functional Protein Design. in (2025).
38. Schymkowitz, J. *et al.* The FoldX web server: an online force field. *Nucleic Acids Res* **33**, W382–8 (2005).
39. Simon, E. & Zou, J. InterPLM: discovering interpretable features in protein language models via sparse autoencoders. *Nat Methods* **22**, 2107–2117 (2025).
40. Gao, L. *et al.* Scaling and evaluating sparse autoencoders. in (2024).
41. Bussmann, B., Leask, P. & Nanda, N. BatchTopK Sparse Autoencoders. Preprint at <https://doi.org/10.48550/arXiv.2412.06410> (2024).
42. van Kempen, M. *et al.* Fast and accurate protein structure search with Foldseek. *Nature Biotechnology* <https://doi.org/10.1038/s41587-023-01773-0> (2023) doi:10.1038/s41587-023-01773-0.
43. Adam Paszke *et al.* PyTorch: an imperative style, high-performance deep learning library. in *Proceedings of the 33rd International Conference on Neural Information Processing Systems* Article 721 (Curran Associates Inc., 2019).
44. Zhong, G., Zhao, Y., Zhuang, D., Chung, W. K. & Shen, Y. PreMode predicts mode-of-action of missense variants by deep graph representation learning of protein sequence and structural context. *Nat Commun* **16**, 7189 (2025).
45. Cheng, J. *et al.* Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* <https://doi.org/10.1126/science.adg7492> (2023) doi:10.1126/science.adg7492.
46. Steinegger, M. & Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat Biotechnol* **35**, 1026–1028 (2017).
47. McGibbon, R. T. *et al.* MDTraj: A Modern Open Library for the Analysis of Molecular Dynamics Trajectories. *Biophys J* **109**, 1528–32 (2015).
48. Tareen, A. & Kinney, J. B. Logomaker: beautiful sequence logos in Python. *Bioinformatics* **36**, 2272–2274 (2020).
49. Huh, M., Cheung, B., Agrawal, P. & Isola, P. Straightening Out the Straight-Through Estimator: Overcoming Optimization Challenges in Vector Quantized Networks. Preprint at <https://doi.org/10.48550/arXiv.2305.08842> (2023).
50. Yin, P. *et al.* Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets. Preprint at <https://doi.org/10.48550/arXiv.1903.05662> (2019).